

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer Examination – 2022

Course: B. Tech. **Branch :** Artificial Intelligence and Data Science **Semester:** IV

Subject Code & Name: BTAIC401 Data Analysis

Max Marks: 60

Date:13/07/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Use of Z-Table, T-Table and F-Distribution table is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Describe the General tips to select samples?	(BT2/CO1)	6
B) Define Missing data? Explain techniques and tools to handle Missing data.	(BT2/CO1)	6
C) Discuss the stages for Data Preparation and Cleaning?	(BT2/CO1)	6
Q.2 Solve Any Two of the following.		12
A) Discuss different types of Kurtosis?	(BT2/CO2)	6
B) i. List and explain Different Measure of Spread or Dispersion with suitable example.		
ii. Identify/Find the Variance and Standard Deviation of the Following Numbers: 1, 3, 5, 5, 6, 7, 9, 10.	(BT1/CO2)	6
C) Describe Statistical Indices with suitable example?	(BT4/CO3)	6
Q. 3 Solve Any Two of the following.		12
A) Define Transformations? Explain Box-Cox Power Transformation?	(BT2/CO3)	6
B) Explain Arcsine Transforms?	(BT2/CO3)	6
C) What is Normal Distribution? Explain in detail?	(BT2/CO3)	6
Q.4 Solve Any Two of the following.		12
A) Apply independent samples T-Test for the following data sets: Data set A: 1,2,2,3,3,4,4,5,5,6 Data set B: 1,2,4,5,5,5,6,6,7,9	(BT3/CO4)	6
B) Explain Chi Square goodness of fit tests using suitable example?	(BT2/CO4)	6
C) Describe McNemar's test with suitable example?	(BT2/CO4)	6
Q. 5 Solve Any Two of the following.		12
A) Describe and explain MANOVA?	(BT3/CO5)	6
B) Differentiate between One way AnovaTest and Two way Anova Test.	(BT4/CO5)	6
C) Discuss Kruskal Wallis ANOVA test with example?	(BT2/CO5)	6

***** End *****